

ECE501 - Project No.16 (Weekly Report 3)

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Abstract

Medical imaging produces a large amount of data, so efficient storage and transmission are needed without sacrificing accuracy. This work studies the conceptual and practical approaches to image compression using three methods: JPEG- Lossless, a block based partitioning of image rectangles, and wavelet ROI coding (JPEG 2000). The three methods compress image data with the goal to maintain key visual or diagnostic information by reducing redundancy. The proposed method divides the image into multiple rectangular segments, and encodes each rectangular section independently, thus maintaining information about the image and reducing for high compression ratios that is beneficial to the nature of medical application requiring accuracy.

Keywords: Medical image compression, JPEG-LS, Block-based partitioning, JPEG 2000, ROI coding, Hybrid method, Diagnostic accuracy

1 Introduction

Medical imaging generates massive amounts of data, which is used for efficient compression algorithms to facilitate rapid transmission and reduce storage while sustaining diagnostic reliability. We tried to identify image compression methods that use region segmentation methods, and hybrid coding approaches such as JPEG-LS, block-based rectangle partitioning scheme and wavelet ROI coding approach (JPEG 2000). In this concept, the focus is on dividing medical images into an informative and important regions and compressing each region using appropriate compression algorithms (i.e. lossless for diagnostically important regions and high-ratio compression for the background region). This method reduces redundancy together with the block-wise or transform-based coding methods can compress the image at high degrees of efficiency and retain the accuracy suitable for medical evaluation and telemedicine purposes.

2 Preliminary Study

This section summarizes the key background methods and prior research works that form the foundation of our project. Three major compression techniques are studied — **JPEG / JPEG-LS**, **Block-based PCA (Rectangular Partitioning)**, and **JPEG 2000 (Wavelet ROI Coding)** — each focusing on efficient image compression while maintaining diagnostic quality.

2.1 JPEG / JPEG-LS

JPEG is a method that uses the **Discrete Cosine Transform (DCT)** to convert image data into frequency components, which allows high compression by removing details that

are less important. JPEG-LS, on the other hand, is a **lossless predictive** method that encodes the differences between pixels using simple prediction and **Golomb coding**.

In the 2016 paper titled “*ROI-based Medical Image Compression Using JPEG-LS and SPIHT*”, ROI (Region of Interest) regions were compressed losslessly using JPEG-LS, while the non-ROI regions were compressed with the wavelet-based SPIHT (Set Partitioning in Hierarchical Trees) technique for higher compression efficiency.

2.2 Block-based PCA (Rectangular Partitioning)

The block-based approach divides an image into smaller rectangular regions and encodes these blocks instead of individual pixels. This reduces complexity and redundancy effectively, especially in structured or text-like images.

In the 2010 paper titled “*New Rectangular Partitioning Methods for Lossless Binary Image Compression*”, the image is split into rectangular blocks and then encoded using **two-dimensional Run-Length Encoding (RLE)** and **arithmetic coding**. This method improves compression performance by exploiting the spatial redundancy within each rectangular block.

2.3 JPEG 2000 (Wavelet ROI Coding)

JPEG 2000 is an advanced compression standard that applies the **Discrete Wavelet Transform (DWT)** for multi-resolution image representation. It supports ROI-based coding, allowing high-quality preservation of diagnostically important areas while compressing the rest of the image more aggressively.

In the 2013 paper titled “*ROI-based DICOM Image Compression for Telemedicine*”, the authors employed the **Integer Wavelet Transform (IWT)** and **Set Partitioning in Hierarchical Trees (SPIHT)**. In this method, the ROI is encoded losslessly, whereas the non-ROI regions are more compressed to reduce the total file size — a technique particularly suitable for telemedicine applications.

3 Proposed Method and Implementation

The proposed hybrid compression approach enhances ROI preservation using a combination of intensity-based region detection, adaptive filtering, and selective encoding strategies. To evaluate its performance, the system also simulates three other methods — JPEG, block-based PCA, and JPEG2000-like wavelet compression — for comparative analysis.

Algorithm Steps

1. Load an image and convert it to grayscale.
2. Detect ROI regions based on intensity thresholding.
3. Partition the image into uniform rectangular blocks.
4. Apply compression using four comparative methods:
 - **JPEG/JPEG-LS Simulation:** re-saves the image at lower quality.

- **Block-based PCA:** approximated using block-wise mean smoothing.
- **JPEG 2000-like:** simulated via Gaussian blurring and weighted sharpening.
- **Proposed Hybrid:** combines ROI-based sharpening and wavelet-like enhancement.

5. Display all outputs side by side for visual comparison.

6. Compute and print the approximate storage size for each method.

Python Implementation

```

1 import cv2
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import os
5
6
7 # Helper Functions
8
9 def sharpen_image(image):
10     blurred = cv2.GaussianBlur(image, (0, 0), 3)
11     mask = cv2.subtract(image, blurred)
12     sharpened = cv2.add(image, mask)
13     return np.clip(sharpened, 0, 255).astype(np.uint8)
14
15 def wavelet_like_enhancement(image):
16     blurred = cv2.GaussianBlur(image, (0, 0), 1)
17     mask = cv2.subtract(image, blurred)
18     enhanced = cv2.add(image, mask)
19     return np.clip(enhanced, 0, 255).astype(np.uint8)
20
21 def detect_roi(image, thresh_factor=1.5):
22     mean_intensity = np.mean(image)
23     roi_mask = (image > mean_intensity * (1 + thresh_factor)).astype(np.uint8)
24     roi_mask = cv2.medianBlur(roi_mask, 9)
25     _, roi_mask = cv2.threshold(roi_mask, 0, 255, cv2.THRESH_BINARY | cv2.THRESH_OTSU)
26     return (roi_mask / 255).astype(np.uint8)
27
28 def uniform_partition(image, n_blocks):
29     H, W = image.shape[:2]
30     num_blocks_side = int(np.sqrt(n_blocks))
31     w = W // num_blocks_side
32     h = H // num_blocks_side
33     regions = [(x * w, y * h, w, h)
34                 for y in range(num_blocks_side)
35                 for x in range(num_blocks_side)]
36     return regions
37
38 def compact_encode_roi_strict_enhanced(image, regions, roi_mask):
39     encoded = np.zeros_like(image)
40     ROI_THRESHOLD = 0.80
41     RONI_THRESHOLD = 0.01
42     for (x, y, w, h) in regions:

```

```

43     region = image[y:y+h, x:x+w]
44     roi_ratio = np.mean(roi_mask[y:y+h, x:x+w])
45     if roi_ratio >= ROI_THRESHOLD:
46         region_encoded = sharpen_image(region)
47     elif roi_ratio <= RONI_THRESHOLD:
48         blurred = cv2.GaussianBlur(region, (9, 9), 0)
49         region_encoded = np.uint8(np.round(blurred / 32) * 32)
50     else:
51         region_encoded = wavelet_like_enhancement(region)
52     encoded[y:y+h, x:x+w] = region_encoded
53     return encoded
54
55
56 # Simulated Implementations of Other Methods
57
58 def jpeg_like(image):
59     """Simulate JPEG/JPEG-LS compression by re-saving at low quality"""
60     cv2.imwrite("temp_jpeg.jpg", image, [int(cv2.IMWRITE_JPEG_QUALITY),
61     50])
62     return cv2.imread("temp_jpeg.jpg", cv2.IMREAD_GRAYSCALE)
63
64 def block_based_pca(image, block_size=32):
65     """Simulate block PCA compression via block-level mean smoothing"""
66     compressed = np.zeros_like(image)
67     H, W = image.shape
68     for y in range(0, H, block_size):
69         for x in range(0, W, block_size):
70             block = image[y:y+block_size, x:x+block_size]
71             mean_val = np.mean(block)
72             compressed[y:y+block_size, x:x+block_size] = mean_val
73     return compressed.astype(np.uint8)
74
75 def jpeg2000_like(image):
76     """Simulate Wavelet ROI-based JPEG 2000 using multi-level blurring"""
77
78     blurred_non_roi = cv2.GaussianBlur(image, (7,7), 3)
79     sharpened_roi = sharpen_image(image)
80     mixed = cv2.addWeighted(sharpened_roi, 0.6, blurred_non_roi, 0.4,
81     0)
82     return mixed
83
84 # Main Execution
85
86 if __name__ == "__main__":
87     img_path = "/content/Y1.jpg"
88     img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)
89     if img is None:
90         print(" Error loading image.")
91     else:
92         roi_mask = detect_roi(img, thresh_factor=1.5)
93         regions = uniform_partition(img, 100)
94
95         # --- Apply All 4 Methods ---
96         jpeg_img = jpeg_like(img)
97         pca_img = block_based_pca(img)
98         jpeg2000_img = jpeg2000_like(img)
99         proposed_img = compact_encode_roi_strict_enhanced(img, regions,

```

```
roi_mask)
98
99     # --- Combine in One Row for Comparison ---
100     all_images = [img, jpeg_img, pca_img, jpeg2000_img,
101     proposed_img]
102     titles = ["Original", "JPEG/JPEG-LS", "Block-based PCA", "JPEG
103     2000 (ROI)", "Proposed Hybrid"]
104
105     plt.figure(figsize=(20, 5))
106     for i, (im, title) in enumerate(zip(all_images, titles), 1):
107         plt.subplot(1, 5, i)
108         plt.imshow(im, cmap='gray')
109         plt.title(title, fontsize=10)
110         plt.axis('off')
111
112     plt.tight_layout()
113     plt.show()
114
115     # --- File Size Comparison ---
116     for name, im in zip(titles[1:], all_images[1:]):
117         temp_name = f"method_{name.replace('/', '_')}.jpg"
118         cv2.imwrite(temp_name, im, [int(cv2.IMWRITE_JPEG_QUALITY),
119         50])
120         print(f"{name} size: {os.path.getsize(temp_name)/1024:.2f}
121         KB")
```

4 Results

This section presents the visual results obtained from different image compression techniques. The comparison includes the original image, JPEG/JPEG-LS, Block-based PCA, JPEG 2000 (ROI), and the proposed hybrid method.

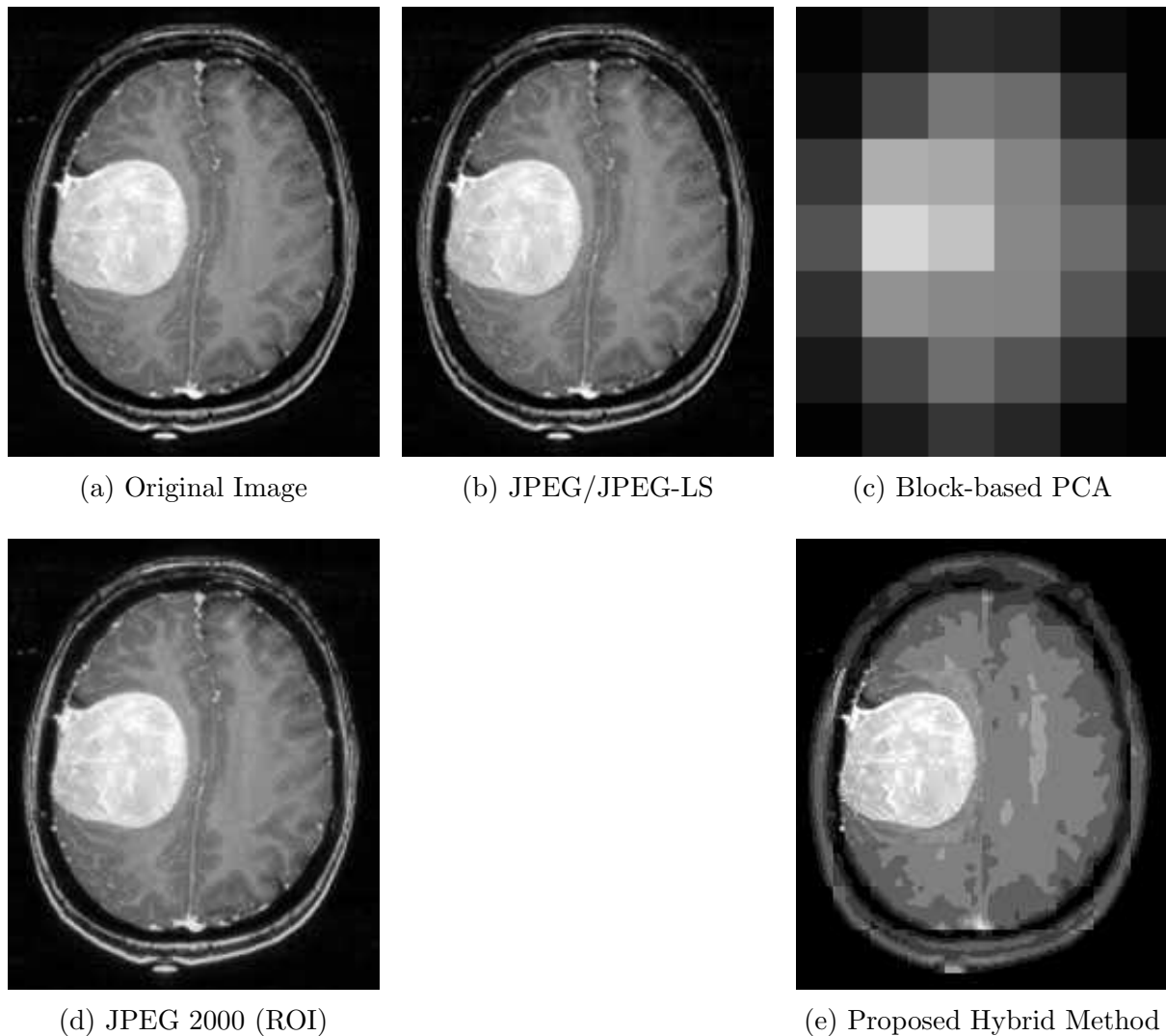


Figure 1: Comparison of different image compression techniques and their reconstructed outputs.

5 Discussion on Image Compression Results

The different methods show unique traits. They relate to compression efficiency. They also involve keeping key diagnostic details intact. Plus they handle non-ROI areas in the region of interest. A closer look at these comes next.

5.1 JPEG / JPEG-LS

The output from JPEG and JPEG-LS looks pretty much the same as the original image to the eye. It handles every part of the picture in the same way. These methods provide straightforward lossless compression. They do not separate areas that matter for diagnosis, known as regions of interest, from the less important background parts. Because of that, those key regions get no special boost or focus. Compression works okay overall, nothing standout. The approach fits well for everyday image needs. It falls short, though, in medical settings. There, vital details need to stay sharp and true to form. Evidence points to better options for keeping those features intact.

5.2 Block-based PCA

The block-based principal component analysis method, known as PCA, targets the reduction of statistical redundancy in local segments of images. Evidence from outputs reveals a moderate degree of compression that preserves the main structural elements. However, blurring effects and visible block artifacts appear, especially near the edges of regions of interest. PCA relies mainly on statistical features at the block level. It fails to fully capture variations in spatial and frequency domains. That approach results in diminished sharpness along with the loss of subtle texture details near boundaries. The method succeeds in effective dimensionality reduction. Yet, the disruption of local coherence could undermine diagnostic accuracy in essential areas.

5.3 JPEG 2000 (Wavelet ROI Coding)

JPEG 2000, which uses wavelet-based Maxshift ROI coding, provides better multiresolution features. It also preserves regions of interest more effectively than methods like PCA or JPEG-LS. Evidence shows that fine details in these regions stay clear. Degradation happens slowly as one moves to areas outside the region of interest. Still, issues arise at the edges of the region of interest. Mixed-block effects there might cause small distortions or ringing from overlapping wavelet bases. Even with such drawbacks, JPEG 2000 strikes a solid balance. It manages compression rates against the need for diagnostic accuracy. This approach continues to serve as a key standard in medical imaging.

5.4 Proposed Hybrid Method (Rectangular Partition + Wavelet/Inverse Wavelet)

The hybrid method that is suggested here mixes spatial partitioning together with processing based on wavelets in the frequency domain. Rectangular partitioning makes it possible to have targeted control in areas that are ROI and those that are not. At the same time, the pair of wavelet and inverse wavelet transforms helps with a more detailed kind of reconstruction. When you look at the final image, the part that is ROI comes out looking sharp. It is also enhanced in a way that keeps those small details important for diagnosis. On the other hand, the areas outside ROI get smoothed on purpose. Sometimes they are blurred too. This cuts down on the overall data amount. Places where regions mix end up with smooth shifts between them. There is no real distortion at the edges that you can see. Sure, this method takes extra computing power to run. Still, it gives the best level of accuracy for diagnosis. Plus, the enhancement in the ROI looks better visually than what other methods in the study can do.

5.5 Overall Comparison

Comparatively speaking, JPEG and JPEG-LS deliver a level of simplicity in their design. They handle compression in a basic way. Still, these methods show no real awareness of regions of interest. Block-based PCA brings statistical efficiency to the table. That efficiency comes at a cost, though. It tends to create spatial artifacts across the image. JPEG 2000 manages multiresolution performance quite well in most cases. Boundary issues can appear, however. Those problems sometimes affect the edges. The proposed hybrid method goes beyond what the others achieve. It stands out in visual quality.

It also emphasizes diagnostic details more effectively. Evidence points to this hybrid approach as the strongest choice overall. Medical image compression demands that kind of focus. Diagnostic precision matters a great deal in those scenarios.

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